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FURTHER STUDIES OF CARBONATE REMAGNETIZATION IN THE NORTHERN APPALACHIAN BASIN

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Abstract. Paleomagnetic and rock magnetic studies of sedimentary carbonates in the northern Appalachian Basin demonstrate that the late Paleozoic remagnetization in this area is probably related to authigenesis of single-domain magnetite. Fifteen sites within Middle Devonian carbonates were sampled along a transect extending from Albany, New York, to the Algonquin Arch in southeastern Ontario. Paleomagnetic studies show that all sites except the two westernmost sites on the Algonquin Arch were remagnetized during Alleghenian time. The very weakly magnetic unremagnetized sites retain a dual-polarity magnetization of probable Devonian age. In the remagnetized sites, both the size of the Kiaman components and magnetite concentration, estimated from low-field susceptibility (X), are highest in the central part of the transect and lower on either side. X ray powder diffraction analyses demonstrate that K-feldspar is also most abundant in the middle of the transect, with lower concentrations to the west and east. We suggest that formation of authigenic magnetite and K-feldspar may be related in this setting. Sites with the largest Kiaman components have the highest unblocking temperatures and the largest ratios of anhysteretic remanent magnetization (ARM) to bulk susceptibility (ARM/X), indicating that these sites have the greatest amounts of single-domain magnetite. Several remagnetized sites exhibit northerly, shallow magnetic components which may be Kiaman "normal" directions. We conclude that remagnetization along the New York-Ontario transect is a chemical remanent magnetization resulting primarily from authigenesis of single-domain magnetite. We present a model in which pulses of heated fluids triggered magnetite formation. Scanning electron microscope observation of magnetic extracts from 14 sites show that most of the resolvable magnetite grains are multidomain grains of irregular shape; many sites also contain a small percentage of magnetic spheres and/or euhedral magnetite particles. A majority of the magnetite grains contain Fe as the only cation, but many also contain minor amounts of other cations including various combinations of Cr, Ni, Ti, Zn, Mn, Si, Al, Ca, and Mg. Single-domain magnetite is not resolvable with our observational techniques, but its presence is inferred from rock magnetic tests. Much of this magnetite may be of authigenic origin.

Introduction

Numerous Paleozoic rocks in North America carry a magnetic component corresponding to the late Paleozoic Kiaman reverse magnetic polarity interval (Kiaman Reversed Superchron). Two types of sedimentary rocks exhibit this late Paleozoic secondary magnetization: red beds and carbonates. In Appalachian Paleozoic red beds, the magnetization resides in hematite and three magnetic components are resolvable: a present field (PF) overprint, a pre-Kiaman (presumably primary) component, and a secondary Kiaman component. Fold tests have shown that the Kiaman magnetization in red beds is a synfolding remagnetization acquired during the late Paleozoic Alleghenian orogeny [Kent and Opdyke, 1985; Miller and Kent, 1986]. In Appalachian Paleozoic carbonates the Kiaman remagnetization usually resides in magnetite and is generally the only magnetization remaining after removal of the overprint of the Earth's present field [e.g., McCabe et al., 1983, 1984]. The carbonate remagnetization also appears to be a synfolding magnetization acquired during the Alleghenian orogeny [McCabe et al., 1983; Bachtadse et al., 1987].

It is generally agreed that the red bed remagnetizations have a chemical origin [Kent and Miller, 1987], but the origin of the remagnetization in the carbonates is still being investigated. Kent [1985] suggests that the late Paleozoic remagnetizations are due to thermoviscous effects. Other workers believe that a chemical remagnetization occurred when diagenetic magnetite formed during the late Paleozoic [e.g., McCabe et al., 1983, 1989; Bachtadse et al., 1987; Suk et al., 1990a,b]. The coincidence of the Alleghenian orogeny and the timing of the Appalachian remagnetizations suggests that there is a relationship between these events. It has been suggested that the formation of diagenetic magnetite was associated with the migration of basinal fluids towards the craton during Alleghenian mountain building events in the late Paleozoic [McCabe et al., 1984].

In addition to Appalachian rocks of the east coast, remagnetization in carbonates has also been reported in the midcontinent region [Lu et al., 1990]. Since it has been suggested that magnetite in remagnetized rocks may be authigenic, it is of interest to note that formation of authigenic minerals has been linked to midcontinent epeirogeny and brine migration. Hay et al. [1988] found that Early Devonian ages for authigenic K-feldspar in upper Mississippi Valley deposits correspond approximately to a major episode of midcontinent deformation. They suggest that K-feldspar formation is related to westward transport of Michigan Basin brine to rocks of the upper Mississippi Valley. It would be of interest to determine whether magnetite concentration is linked with K-feldspar and other authigenic minerals in these remagnetized rocks. Magnetite concentration in remagnetized carbonates of the Onondaga Formation in New York

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has been linked with percent illite (in illite-smectite mixed layer clays) in a bentonite layer within the Onondaga Formation [Jackson et al., 1988], and K/Ar dating of other K-bentonites in the Appalachian Basin demonstrate Alleghenian ages [Elliott and Aronson, 1987]. In addition, Suk et al. [1990b] found a close association between dolomite, K-feldspar, quartz, and authigenic magnetite in Ordovician carbonates from eastern Tennessee. These observations suggest a correlation between remagnetization and synthesis of diagenetic or authigenic minerals.

This paper will focus on the origin of the remagnetizations in carbonates of the northern Appalachian Basin. This study is a continuation of the work of McCabe et al. [1989], who concluded that remagnetization in carbonates of the southern Appalachian Basin is related to magnetite authigenesis and that the remagnetization in that setting must be due to chemical processes.

Geological Setting and Sampling

Oriented samples were collected from 15 sites along a transect extending from Albany, New York, to the Algonquin Arch in southeastern Ontario (Figure 1). Along this transect, Middle Devonian limestones of the Onondaga Formation in New York grade laterally into limestones and dolostones of the equivalent Detroit River Group in Canada.

Since the causes of remagnetization during the late Paleozoic are being studied, the temperature profile across the northern Appalachian Basin during this time is of interest. Clay-mineral diagenesis and apatite fission track data obtained from the Middle Devonian Tioga metabentonite (within the Onondaga Formation) indicate that maximum paleotemperatures rise from low values in western New York state to higher values in the east [Johnsson, 1986]. Various geothermal indicators, including fluid inclusions, vitrinite reflectance, and conodont color alteration indicate a temperature gradient ranging from about 100°C near Buffalo to about 200°C near Albany [Friedman and Sanders, 1982; Friedman, 1987; Epstein et al., 1977]. No thermal maturity data are available for the Canadian part of the profile, but a continued westward decrease in late Paleozoic paleotemperatures would be expected towards the Algonquin Arch. High temperatures in the northern Appalachian Basin are attributed to burial heating during Alleghenian time, with maximum burial depth estimated to be from 6.5 to 7 km [Friedman and Sanders, 1982; Johnsson, 1986; Friedman, 1987]. Great burial depth of northern Appalachian sediments during the late Paleozoic implies great post-Paleozoic uplift and erosion to bring these formerly deeply buried strata to the present land surface. Apatite fission track ages indicate that the northern Appalachian Basin was uplifted during the Mesozoic Era [Johnsson, 1986].

Methods

Paleomagnetic Methods

Paleomagnetic samples were processed in the paleomagnetic laboratory at Louisiana State University. Most oriented samples were demagnetized using stepwise thermal demagnetization (TD). A few samples from each site were subjected to stepwise alternating field (AF) demagnetization. TDs were performed using a three-zone furnace constructed from Lindberg components and contained in three nested mu-metal shields. The residual field is less than 50 nT in the furnace and less than 5 nT in the cooling chamber. AF demagnetization

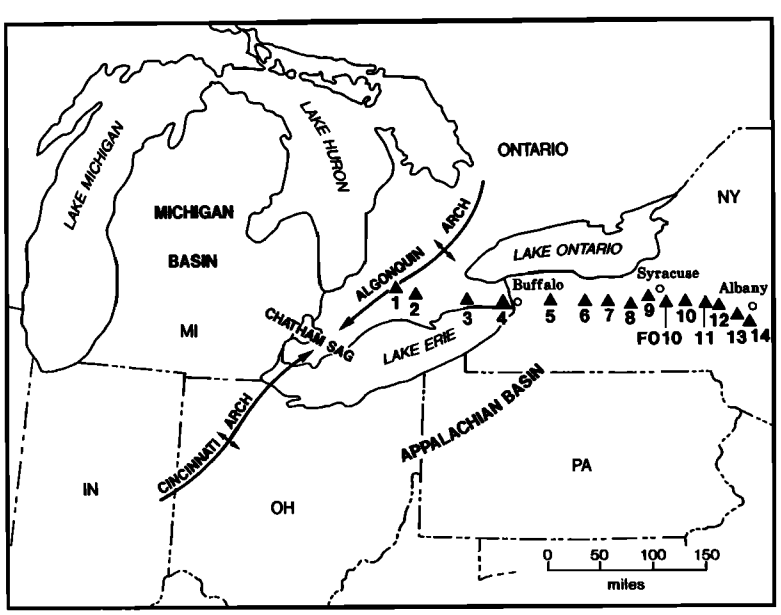


Fig. 1. Map showing the location of sampling sites for the New York-Ontario transect. Solid triangles represent sites from which Devonian samples were collected.

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was carried out using a shielded coil tuned to 653 Hz and driven by a Behlman oscillator and AC power supply. Remanent magnetization was measured in a cryocooled, three-axis superconducting magnetometer manufactured by CTF Systems. Demagnetization data were plotted on orthogonal demagnetization diagrams [Zijderveld, 1967], and the directions of the component magnetizations were determined using principal component analysis [Kirschvink, 1980].

After the present field and Kiaman magnetization directions were determined for a specimen, computer analysis was used to generate a model unblocking temperature (T_{ub}) spectrum for estimating the magnitudes and T_{ub} of the component magnetizations. This was accomplished by projecting difference vectors onto the plane defined by the PF and ancient magnetization directions, and resolving each projection into a component in the PF direction and a component in the Kiaman field direction. The model data were then plotted on histograms (Figure 2), and the magnitude and T_{ub} range of each component magnetization were estimated. This method assumes only that (1) each specimen's natural remanent magnetization (NRM) is composed of only two component magnetizations, the directions of which are known, and (2) any deviation of a difference vector from the plane defined by the two end-member magnetizations is the result of random noise and can be ignored. Generally, it is unknown whether these assumptions are valid, but in the case of these remagnetized carbonates, the two-component nature of the NRM has been confirmed in numerous investigations.

Rock Magnetic Methods

In order to identify the minerals carrying the magnetic remanence and to estimate the relative concentration and size distribution of magnetite across the transect, a number of rock magnetic tests were performed. The mineralogy of the magnetic carriers was determined by acquisition and TD of isothermal remanent magnetization (IRM) [Dunlop, 1972; Lowrie and Heller, 1982]. Relative concentrations and grain sizes of magnetite were estimated using NRM, IRM (at 0.3 T), X, and ARM [King et al., 1982]. IRMs were imparted with an impulse magnetizer, in fields up to 1.1 T. X was measured with a Sapphire Instruments susceptibility bridge, and remanence measurements were made on the CTF instrument at Louisiana State University. ARMs were imparted in the AF device described above, with a DC bias field of 0.1 mT.

Magnetic Extracts

Magnetic extracts were prepared to observe grain morphology and to qualitatively determine the elemental composition of the magnetic carriers. Observations were made using a scanning electron microscope (SEM) equipped with an energy-dispersive X ray spectrometer (EDAX). Magnetic extracts for each site were obtained by dissolving the crushed rock sample in dilute buffered acetic acid [McCabe et al., 1983] and slowly passing the dissolved sample over a rare earth permanent magnet several times.

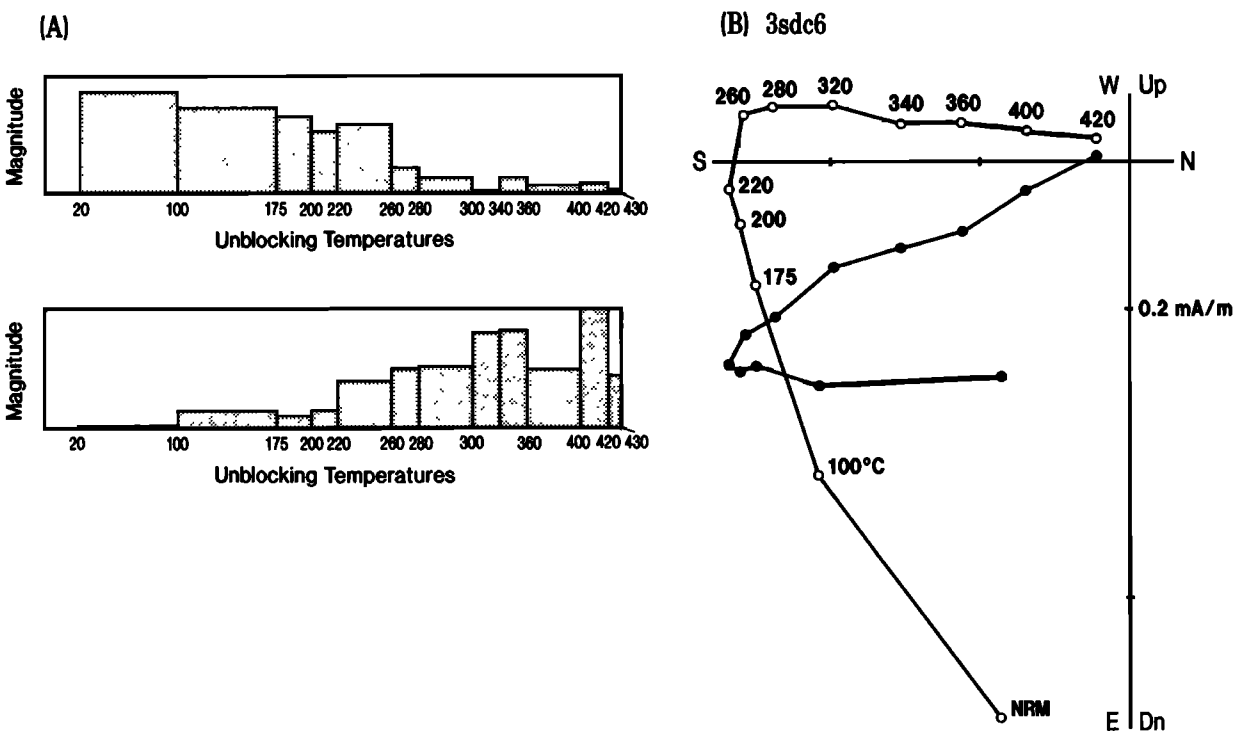


Fig. 2. (a) Histograms showing magnitudes and unblocking temperature ranges of the PF (upper) and Kiaman (lower) magnetization vector components for sample 3sdc6. (b) Zijderveld plot generated from thermal demagnetization data, showing the magnetization vectors used to produce the histograms.

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X Ray Diffraction

In order to determine the relative concentrations of various minerals across the transect, X ray powder diffraction was performed on samples from each site using an automated Phillips diffractometer and Cu k-alpha radiation. Computer analysis of the X ray diffraction data was used to estimate the concentrations of the mineral phases present. The method used by the computer for analysis of the diffraction data is briefly summarized as follows: (1) smooth the digital pattern and subtract the background, then determine the location and height of all diffraction peaks; (2) determine the presence of quartz or calcite and correct the location of the peaks using one of these minerals as an alignment standard; (3) identify the minerals present on the basis of a single diagnostic peak and subtract interfering secondary peaks of other minerals from the diagnostic peak; (4) multiply all corrected diagnostic peak heights by their intensity factors (determined from standards) to convert them into mineral ratios; and (5) adjust the total of all identified minerals to 100% [Cook et al., 1975]. Computer analyses of the X ray diffraction results were used to estimate the percentage of various minerals, including aluminosilicates such as K-feldspar, illite, and other clays for each site.

Results

Kent [1979, 1985] and McCabe et al. [1989] demonstrated that the Devonian Onondaga Limestone was remagnetized during the Kiaman reverse magnetic polarity interval. The mineral carrying the remagnetization was shown to be magnetite. Most of the sites we sampled across upstate New York and into southeastern Ontario have a clearly resolvable Kiaman magnetization. However, sites 1 and 2 on the Algonquin Arch, the westernmost sites sampled (Figure 1), demonstrate no Kiaman remagnetization. Results from the remagnetized and unremagnetized sites will be presented separately.

Unremagnetized Sites

Paleomagnetism. Sites 1 and 2 on the Algonquin Arch have the weakest NRMs observed in this study (Figure 3) and no discernable Kiaman components. A number of samples from these sites show either north, up or south, down magnetic directions (Table 1 and Figure 4), which appear to be dual-polarity Devonian magnetizations. The paleomagnetic poles for the characteristic directions of sites 1 and 2 are 22.4°N, 98.0°E and 28.8°N, 93.9°E respectively, corresponding with North American poles of Devonian age (reviewed by Van der Voo [1989]). The north, up and south, down components

TABLE 1. Summary of Site Paleomagnetic Data

	Location		Mean PF Direction (D/I), deg	N/N_o	k	a_{95} deg	Mean Kiaman Direction (D/I), deg	N/N_o
	Latitude °N	Longitude °W						
Site								
1	43.2	81.1	---	---	---	---	---	---
2	43.0	80.8	343.3/66.5	5/26	26.0	15.3	---	---
3	42.8	79.9	357.6/66.6	25/28	29.4	5.4	154.2/-9.1	27/28
4	42.8	79.2	354.7/60.4	23/28	31.6	5.5	160.6/-3.0	14/28
5	42.9	78.1	16.1/56.8	29/33	22.5	5.8	193.9/-17.0*	1/33
6	43.0	77.6	357.4/63.9	37/44	23.8	4.9	168.8/8.9	7/44
7	43.0	77.2	347.2/66.9	24/24	83.4	3.3	163.9/-5.7	17/24
8	42.8	76.8	11.4/60.0	23/28	30.9	5.5	157.0/-14.3	22/28
9	43.0	76.3	345.2/69.1	35/35	53.5	3.3	164.5/-9.7	35/35
FO10	43.0	75.7	351.1/65.5	27/29	68.6	3.4	157.9/1.6	16/29
10	43.0	75.6	356.9/63.8	19/22	203.0	2.4	151.2/-0.5	16/22
11	42.9	75.0	345.8/58.8	26/26	88.1	3.0	173.2/-5.1	4/26
12	42.8	74.7	358.9/62.5	20/20	285.8	1.9	161.3/-2.0	19/20
13	42.7	74.1	344.1/68.7	31/33	223.6	1.7	160.3/-9.2	33/33
14	42.5	73.9	341.8/59.7	21/44	19.4	7.4	166.3/-4.8	20/44
Means			354.3/63.9				161.6/-4.5	
k			172.2				87.4	
K								
a_{95}			3.0				4.7	
A_{95}								

* Omitted from mean.

**The directions recorded for sites 1 and 2 are primary Devonian directions. S, down directions were transposed to N, up to compute site means. These Devonian directions were omitted from the overall mean at the bottom of the column.

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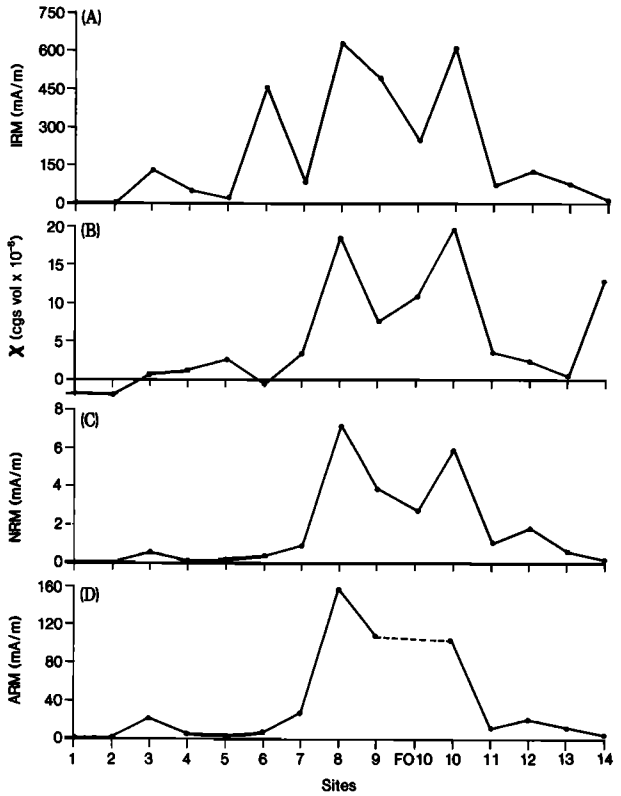


Fig. 3. Comparison of average magnetic intensities across the transect, showing the correlation among intensity values and demonstrating that intensities are greatest in the center of the study area. (a) IRM, (b) X, (c) NRM, and (d) ARM. Dashed portion of the curve in ARM graph represents missing value.

generally have low unblocking temperatures, ranging from 245 to 295°C, such as PF components have in other samples. However, in samples which exhibit north, up or south, down directions, there is no PF overprint. The Devonian directions are the only resolvable directions. Possibly, the Devonian component is carried by small, relatively stable magnetic grains. This type of magnetization might also be present in the remagnetized sites but may be unresolvable because it is swamped by the magnetizations associated with late Paleozoic magnetite authigenesis.

The absence of a PF magnetization in the majority of samples from sites 1 and 2 is interesting. No samples from site 1 and only five samples from site 2 have a clearly resolved PF component (Table 1). The absence of a PF component may be due to the absence of a significant population of large, unstable multidomain grains. In fact, about half of the samples from sites 1 and 2 have no resolvable

Showing PF, Kiaman, and Northerly Directions

k	a_{95}, deg	Kiaman Pole		Mean Northerly Direction (D/I),deg	N/N_o	k	a_{95}, deg	Pole	
		Latitude °N	Longitude °E					Latitude °N	Longitude °E
---	---	---	---	1.0/-42.2**	10/26	7.4	19.1	---	---
---	---	---	---	4.9/-33.0**	3/26	18.1	29.9	---	---
82.7	3.1	45.4	139.1	---	---	---	---	---	---
15.5	10.4	44.9	131.8	---	---	---	---	---	---
---	---	---	---	6.8/2.8	5/33	15.1	20.4	-48.1	271.6
45.0	9.1	41.6	118.9	345.1/6.3	3/44	21.8	27.1	-48.1	304.9
81.9	4.0	47.6	127.4	353.2/-23.5	5/24	9.8	25.8	-34.4	290.9
41.0	4.9	49.2	138.4	---	---	---	---	---	---
414.3	1.2	49.1	127.7	---	---	---	---	---	---
36.4	6.2	44.2	132.0	---	---	---	---	---	---
75.8	4.3	40.5	142.6	---	---	---	---	---	---
14.2	25.2	49.2	115.4	3.4/-4.6	4/26	7.1	37.0	-44.7	280.3
41.2	5.3	44.2	132.5	---	---	---	---	---	---
233.9	1.6	46.6	135.7	---	---	---	---	---	---
13.7	9.2	50.5	131.8	357.2/6.2	4/44	6.7	38.3	-50.5	290.5
		46.3/131.2		357.2/-2.5				-45.7/287.7	
		161.9		28.8				57.5	
		3.4		14.5				10.2	

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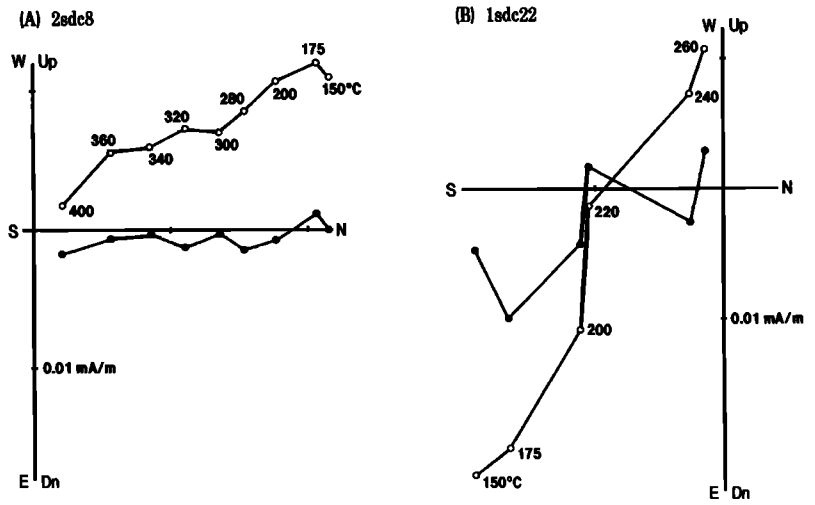


Fig. 4. Plots of thermal demagnetization data showing (a) north, up and (b) south, down directions seen in samples from sites 1 and 2.

components at all, probably because of their extremely weak magnetization.

Rock magnetism. Sites 1 and 2 are the most weakly magnetic sites of our transect (Figure 3), indicating a very low total concentration of magnetic minerals. During stepwise acquisition of IRM, saturation generally occurred by 0.3 T, indicating that magnetite is probably the dominant magnetic phase. To determine the blocking temperatures of the magnetic carriers, two procedures were used. An IRM was imparted at 1.1 T along the -V sample axis, or a composite IRM was imparted at 1.1 T along the -V sample axis and at 0.2 T along the +V axis [Channell et al., 1982]. The samples were then thermally demagnetized. During stepwise TD of both the single and composite IRMs, magnetization was removed continuously from low-coercivity grains up to 540-580°C in most samples. This is further evidence that magnetite is the dominant magnetic mineral.

Magnetic extracts. There was not sufficient extract from site 1 to observe with the SEM. The extract from site 2 contains many irregularly shaped particles, some of which appear to have high-temperature oxidation exsolution lamellae. About 10% of these particles contain Fe as the only cation while the other 90% contain Fe plus minor amounts of other cations, such as Cr, Ni, Al, and Si. At least some of these magnetites are probably detrital [Freeman, 1986]. Also observed were a small number of spherical particles which contain only Fe cations and a fair number of euhedral particles. Half of the euhedral particles contain only Fe cations, while the other half contain Fe as well as a small amount of another cation, such as Cr or Ti (Table 2).

Remagnetized Sites

Paleomagnetism and rock magnetism. Sites 3-14 all exhibit Kiaman (southeast, shallow) directions of magnetization. At least some samples from each of these sites retain resolvable Kiaman components, and for most of these sites the majority of samples exhibit a Kiaman direction (Table 1).

Intensity measurements, including IRM, ARM, and NRM, demonstrate that site 3, near the base of the Algonquin Arch (Figure 1), has moderate magnetic intensity (Figure 3). Even so, it is substantially more magnetic than the areas around it (i.e., sites 1, 2, 4, 5, and 6). Almost every sample from site 3 has a well-resolved PF and Kiaman component (Table 1). In fact, of the sites sampled in this study, sites 3, 9, and 13 (Figure 1), with moderate to high magnetic intensities (Figure 3), have the best resolved and most tightly clustered Kiaman directions as well as clearly resolved PF components (Table 1 and Figure 5). The Kiaman components at these sites generally go to the origin (Figure 5) and have fairly high average maximum T_{ub} ranging from 445°C at site 3 to 497°C at site 9 to 502°C at site 13. The high T_{ub} at these sites correspond with large ARM/X ratios (Figure 6), indicating that higher T_{ub} are associated with smaller average grain size [Banerjee, 1981]. Grain size will be discussed in more detail later.

The samples from between sites 3, 9, and 13 demonstrate more variable magnetic components. Sites 4, 5, and 6 (Figure 1) all have clear PF components, but the Kiaman directions are often poorly resolved. Site 4, with fairly low magnetic intensity (Figure 3) has well-resolved Kiaman components in about half the samples, while sites 5 and 6 have resolvable Kiaman directions in only a few samples (Table 1). Site 5 has low moderate magnetic intensity and site 6, whose average NRM intensity is low, has surprisingly high values for the IRM in many samples (Figure 3). Although all samples from site 6 have fairly low NRM intensities, those which have somewhat higher NRM intensities generally have high IRM intensities and vice versa. Using rock magnetic tests, it was found that most samples with high IRM intensities contain significant amounts of pyrrhotite in addition to magnetite; those samples with low IRM intensities generally contain little or no

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TABLE 2. Summary of Particles Found in Magnetic Extracts

Site	Particle Type	Size Range, (microns)	Cations
1	insufficient extract		
2	irregular	7-37	Fe,Cr,Ni,Al,Si
	euhedral	10-30	Fe; Fe(Cr,Ti)*
	spheres	---	Fe
3	irregular	13.5-23.5	Fe; Fe(Cr,Ni,Ti,Ca,Al,Si,Mg)*
	euhedral	33.5	Fe
	spheres	---	Fe; Fe,Zn; Fe,Cr; Fe,Mn
4	irregular	19-53.5	Fe; Fe(Cr,Ni,Mn,Si,Al);* Cr,Fe,Zn
	spheres	10-16	Fe; Fe,Mn; S,Fe; Cr,Mn
5	aggregates	30	Fe; Fe(Cr,Si,Al,Ca)*
	irregular	40-70	Fe; Fe(Ti,Cr,Ni,Mn,Al,Si)*
	spheres	2-46	Fe; Fe,Ti; Fe,Zn; Fe,Ca; Fe,Ti,Si,Al; Fe,Ca,Zn; Fe,Ca,Al,Si
6	aggregates	30-180	Fe; Fe,S(Cr,Si,Al,K,Mg)*
	irregular	10-20	Fe
	euhedral	30	Fe
7	irregular	6-120	Fe; Fe(Ti,Cr,Ni,Mn,Zn,Si,Al,Mg,K,Ca)*
	euhedral	37.5	Fe
	spheres	2-16	Fe
8	irregular	20.5-33.5	Fe; Fe,Cr; Fe(Ca,P,Si,Al)*
	euhedral	10-33.5	Fe; Fe,Cr; Fe,Al,Si
	spheres	<1-10	Fe; Fe,Cr; Fe,Al; Fe,Mn
9	irregular	12-80	Fe; Fe(Ti,Cr);* Fe,Si,Al; Fe,P,S,Si,Al,Ca
	euhedral	23	Fe
	spheres	10	Fe
FO10	aggregates	60	Fe
	irregular	24-66	Fe
	spheres	---	Fe
10	aggregates	25	Fe
	irregular	12-18	Fe; Fe,Cr; Fe,Si,Al
11	irregular	30-100	Fe; Fe,Cr; Fe,Ni; Fe,Si,Al; Ti,Si,Al,Mg,Ca
	hollow rings	4-24	Fe
12	irregular	25-53	Fe; Fe(Ni,Cr,Al,Si)*
	spheres	12	Fe; Fe,Cr
13	aggregates	---	---
	irregular	13.5-80	Fe; Fe(Cr,Mn,Al,Si)*
	flat	---	Fe
	spheres	11.5	Fe
14	irregular	23	Fe; Fe(Cr,Mn,Al,Si)*
	flat	---	Fe; Fe,Cr
	spheres	5-22	Fe

This table contains only qualitative descriptions of the extracts. Sizes and cation compositions apply to a limited number of grains tested.

*Cations are present in various combinations in different grains.

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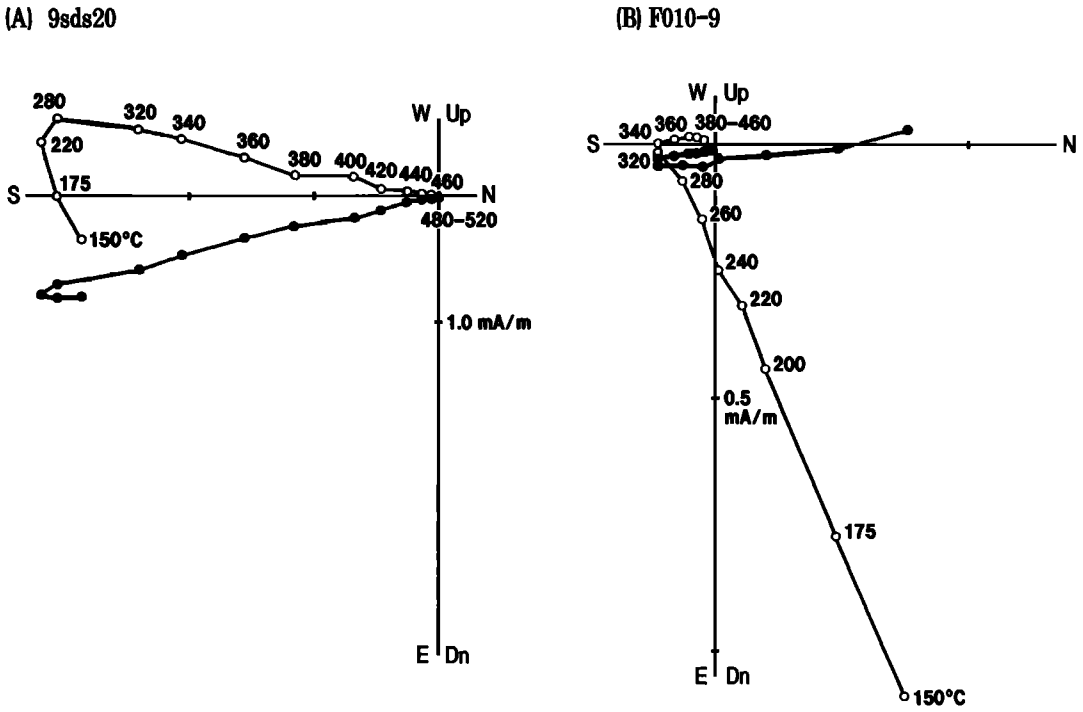


Fig. 5. Plots of thermal demagnetization data. (a) Large, well-resolved Kiaman component such as that seen in samples from sites 3, 9, and 13. (b) Small Kiaman component such as that seen in samples from sites F010, 10, 11, and 12.

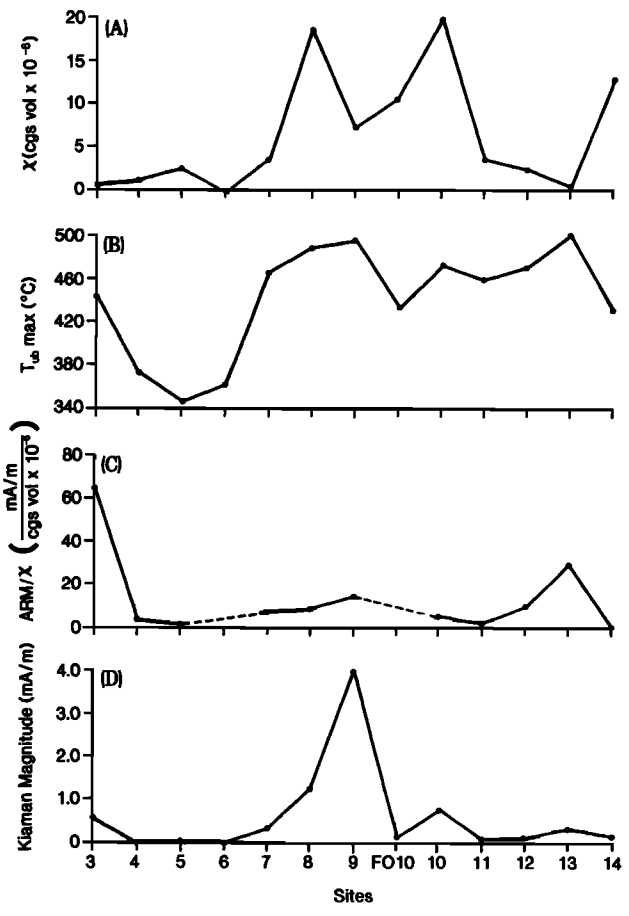


Fig. 6. Comparison of various parameters for remagnetized sites 3-14. (a) Average values for X. (b) Average maximum unblocking temperatures of the Kiaman components. (c) Ratios of ARM/X. Dashed lines represent negative average X value (site 6) or missing ARM data (site F010). (d) Average sizes of the Kiaman (south, shallow) components.

pyrrhotite. Pyrrhotite was identified by stepwise TD of IRM, which demonstrated a rapid thermal decay in the range 300-350°C (Figure 7) [Lowrie, 1990]. In addition, minerals containing Fe and S (assumed to be pyrrhotite) were observed in magnetic extracts from site 6 (Figure 8). It is concluded, therefore, that the high IRM values at site 6 are related to the presence of pyrrhotite. Pyrrhotite, however, probably does not carry the Kiaman magnetization at site 6. It was found that most samples from site 6 with resolvable Kiaman components have low intensity values, indicating that the Kiaman remagnetization resides in magnetite rather than pyrrhotite.

The poor Kiaman directions and intensity characteristics at sites 5 and 6 may be related to their composite mineralogy. As discussed above, site 6 has a large amount of pyrrhotite, and site 5 contains significant amounts of the weakly magnetic mineral hematite (Table 3). Possibly, these minerals formed during post-Kiaman time and deluged or wiped out the Kiaman magnetizations in these areas.

Interestingly, sites 5 and 6 (as well as site 7) appear to have north, shallow directions in a

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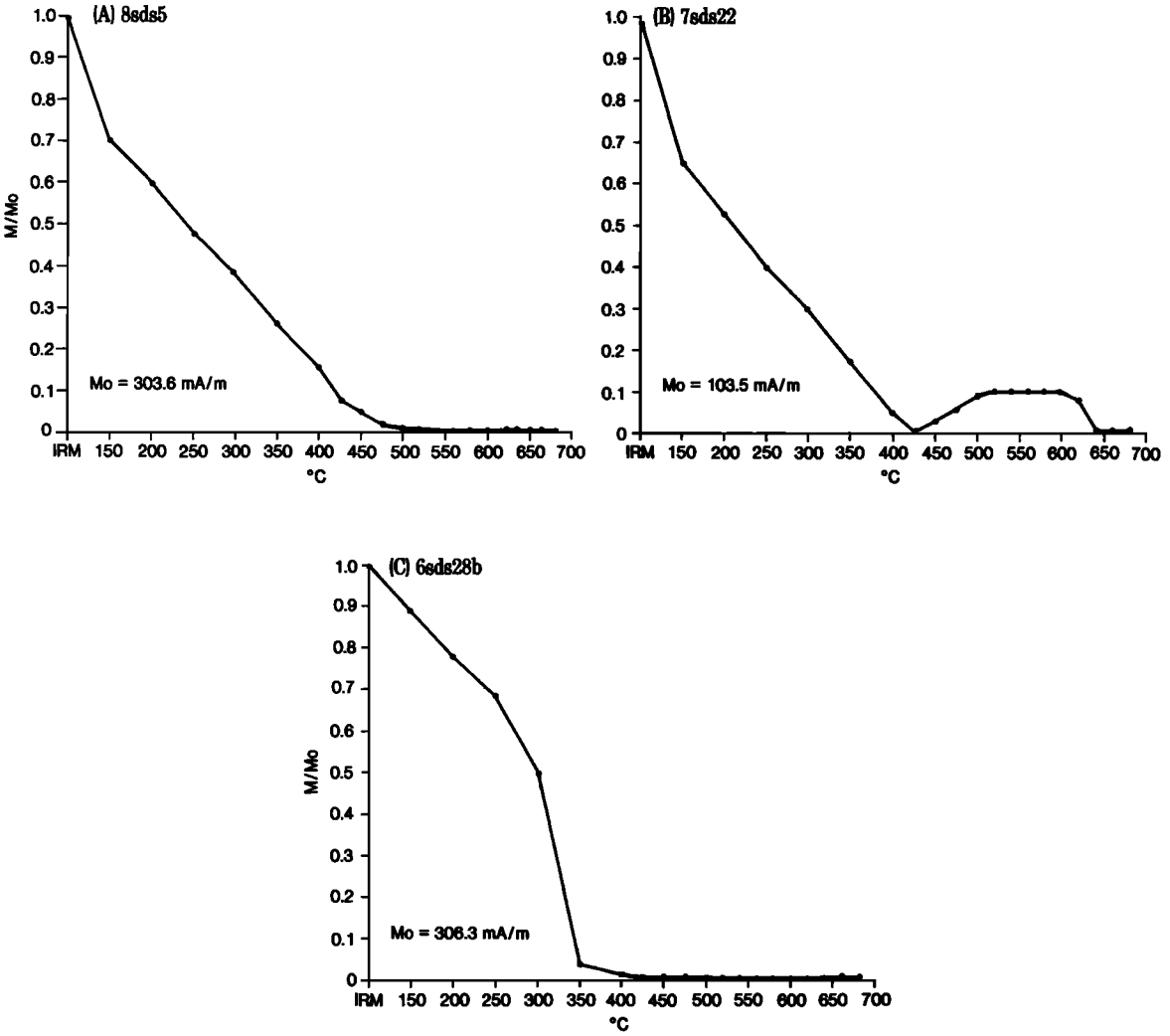


Fig. 7. Thermal demagnetization of composite IRM using the method of Channell et al. [1982] showing the presence of (a) magnetite, (b) hematite, and (c) pyrrhotite in samples from different sites.

number of samples (Table 1). These may be Kiaman normal directions. In general, the solitary PF components in samples from sites 4, 5, and 6 and the north, shallow components (when present) in samples from sites 5, 6, and 7 do not go to the origin; they either fall short of the origin or bypass the origin and approach a south, shallow direction (Figure 9). Some of these samples may have unresolved south, shallow Kiaman components as well as PF directions (and sometimes north, shallow Kiaman components) of magnetization. Sites 4, 5, and 6 all apparently have low maximum T_{ub} for the Kiaman directions (Figure 6). However, it may be that the Kiaman component becomes unresolvable due to the noise at higher temperatures, leading to underestimation of the T_{ub} at these sites.

Sites 7, 8, 9, FO10, and 10 in the central part of the transect (Figure 1) all have well-resolved PF and Kiaman components (Table 1), although there is variable magnetic intensity among these sites. Intensity of magnetization ranges from moderate at site 7 to fairly high at site FO10 to high at sites 8, 9, and 10 (Figure 3). Sites 7-10 extend west to east from Clifton Springs to Peterboro, New York. Of all sites across the transect, site 8 has the greatest magnetic intensity (Figure 3), and site 9 has the best resolved Kiaman components (Table 1 and Figure 5). These results suggest that the central part of the transect may have undergone the most intensive remagnetization during the Kiaman interval, resulting in moderate to high intensities of magnetism (Figure 3) and very clearly resolved Kiaman directions over the entire area.

The area east of site 10, which includes sites 11, 12, 13, and 14 (Figure 1), again demonstrates more variable results with respect to both intensity of magnetization and magnetic directions. Site 11, which has low moderate magnetic intensity (Figure 3), has well-resolved PF directions but poorly resolved or unresolvable Kiaman components in most samples. A few samples from site 11 apparently have north, shallow magnetic components (Table 1), similar to those seen in sites 5, 6, and 7. Again, these north, shallow components might be Kiaman normal directions. As seen in other sites with poor Kiaman directions, the PF and

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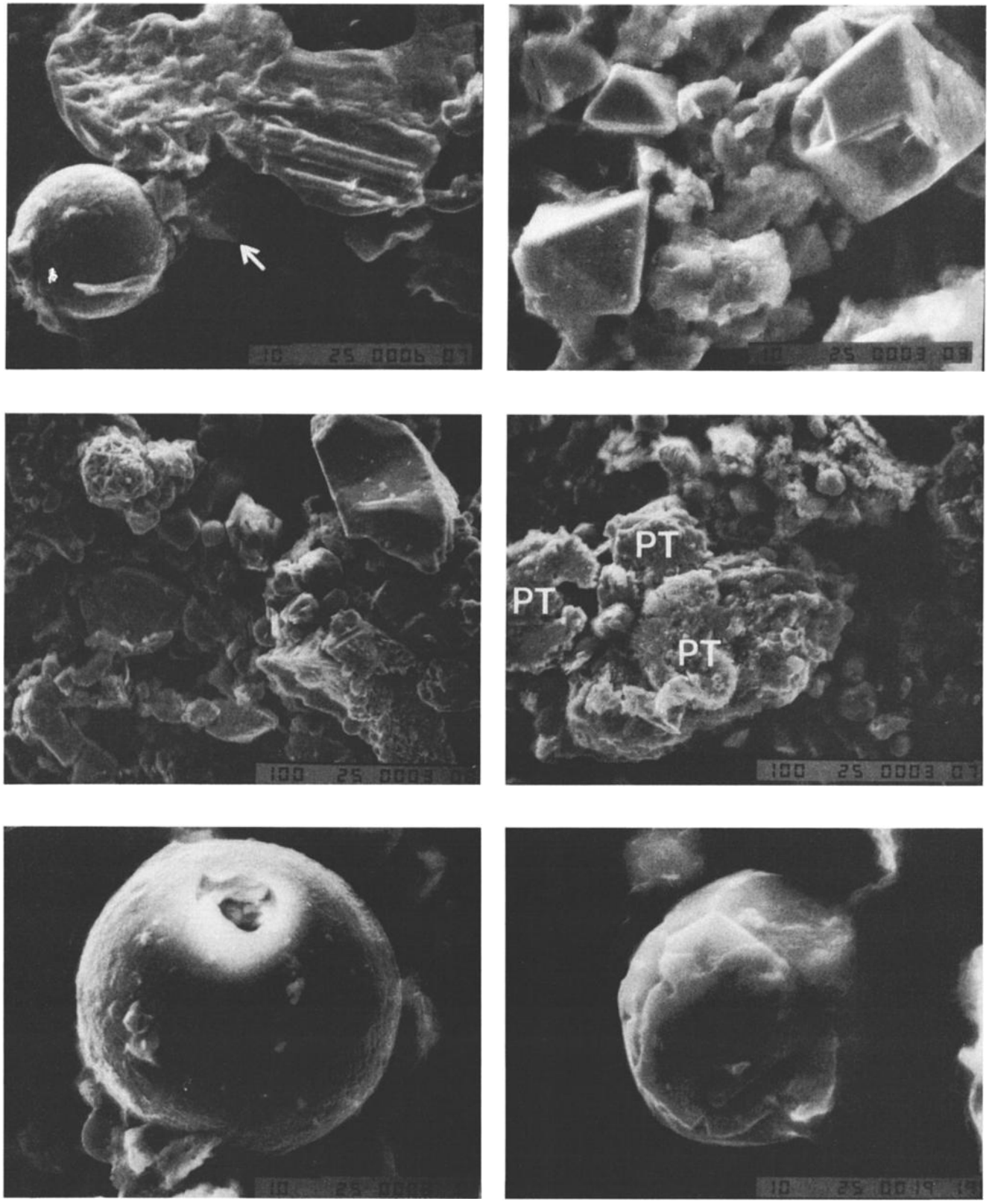


Fig. 8. Scanning electron micrographs of magnetic extracts from different sites. (a) Site 2. Large, irregular grain, euhedral grain (arrow) and sphere. (b) Site 8. Euhedral grains and irregular grains. (c) Site 7. Irregular grains and a few small spheres. (d) Site 6. Large grains are pyrrhotite (PT); small grains in the background are magnetite. (e) Site 5. Large and small spheres. Large sphere contains Fe, Ti, Si, and Al cations. (f) Site 7. Sphere composed of fused plates. (g) Site 14. Large hollow sphere and small sphere.

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Fig. 8. (continued)

north, shallow components do not go to the origin but generally approach a south, shallow direction (Figure 9). These samples may have unresolvable south, shallow Kiaman directions.

On the eastern end of the transect, sites 12 and 13, with moderate and low moderate magnetic intensities, respectively, have clear PF and Kiaman components in almost all samples (Table 1). In fact, as previously explained, site 13 has exceptionally clear and well-clustered Kiaman directions, although the magnetic intensity at this site is low moderate (Figure 3).

Site 14, the easternmost site sampled (Figure 1), differs from the other sites along the transect. While sites 1-13 are all flat-lying carbonate rocks, site 14 is a slightly tilted clay shale containing about 46% quartz. Samples from site 14 have low magnetic intensities (except for X) (Figure 3), and most have fairly well-resolved PF and Kiaman directions (Table 1). However, many samples from this site seem to have meaningless magnetic directions (e.g., southwest, up; north, steeply up), indicating that some areas within site 14 are magnetically unstable. Site 14, like sites 5, 6, 7, and 11, exhibits north, shallow components in a number of samples (Table 1). As described previously, these north, shallow directions do not go to the origin but approach a south, shallow direction (Figure 9).

The average sizes of the Kiaman components at different sites across the transect are very variable (Figures 5, 6, and 10). An interesting observation with respect to this is that some samples from site 10 and most samples from sites FO10, 11, and 12, (in the eastern half of the study area (Figure 1)) have extremely small Kiaman components compared to other sites with well-resolved Kiaman directions. This is especially clear in the graph of Kiaman component/total magnetic components (Figure 10). This graph also demonstrates that sites 3, 9, and 13, in the western, central, and eastern part of the transect, respectively (Figure 1), have comparatively large Kiaman components.

Magnetic carriers. In order to determine the identity of the magnetic carriers across the transect, various methods were used (as described previously). Using these methods, magnetite was found to be the dominant magnetic carrier at most sites, although some sites contain significant amounts of hematite or pyrrhotite (Figures 7 and 11 and Table 3). In sites 3, 4, 8, 9, FO10, 10, 11, 13, and 14 (Figure 1), magnetite is the dominant magnetic mineral, although traces of hematite and/or pyrrhotite are frequently present (Table 3). Sites 5, 7, and 12 (Figure 1) contain substantial amounts of both magnetite and hematite (Figures 7 and 11 and Table 3). The mineralogy of site 6 (Figure 1) differs from the others along the transect. Although some samples from this site contain magnetite as the dominant magnetic carrier, the majority of samples from site 6 contain both magnetite and pyrrhotite in significant quantities (Figures 7 and 8 and Table 3).

The presence and resolution of magnetic directions may be related to the combination of magnetic minerals at a particular site, but this is probably not the ultimate controlling factor. Areas with all of the mineralogies described above contain PF and Kiaman magnetic components, although these components may differ in size and

TABLE 3. Summary of magnetic minerals across the transect

Site	Major Minerals	Minor Minerals
1	magnetite	hematite
2	magnetite	hematite; pyrrhotite
3	magnetite	pyrrhotite
4	magnetite	pyrrhotite
5	magnetite; hematite	pyrrhotite; goethite
6	magnetite; pyrrhotite	hematite
7	magnetite; hematite	pyrrhotite
8	magnetite	hematite; pyrrhotite
9	magnetite	pyrrhotite
FO10	magnetite	pyrrhotite
10	magnetite	---
11	magnetite	---
12	magnetite; hematite	pyrrhotite; goethite
13	magnetite	pyrrhotite
14	magnetite	pyrrhotite

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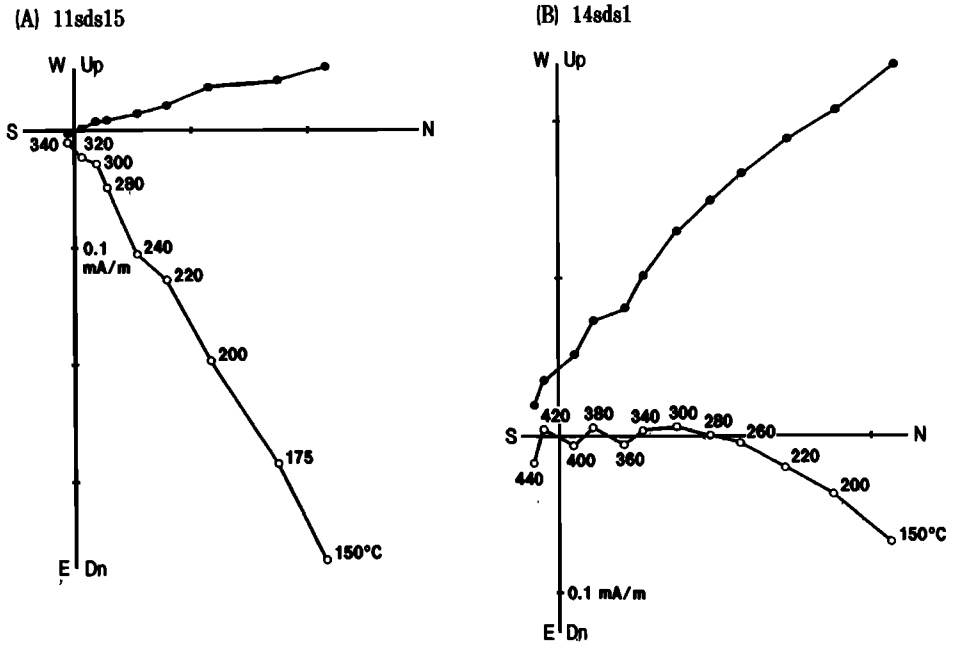


Fig. 9. Plots of thermal demagnetization data showing (a) PF and (b) north, shallow components which do not go to the origin but instead approach a south, shallow direction.

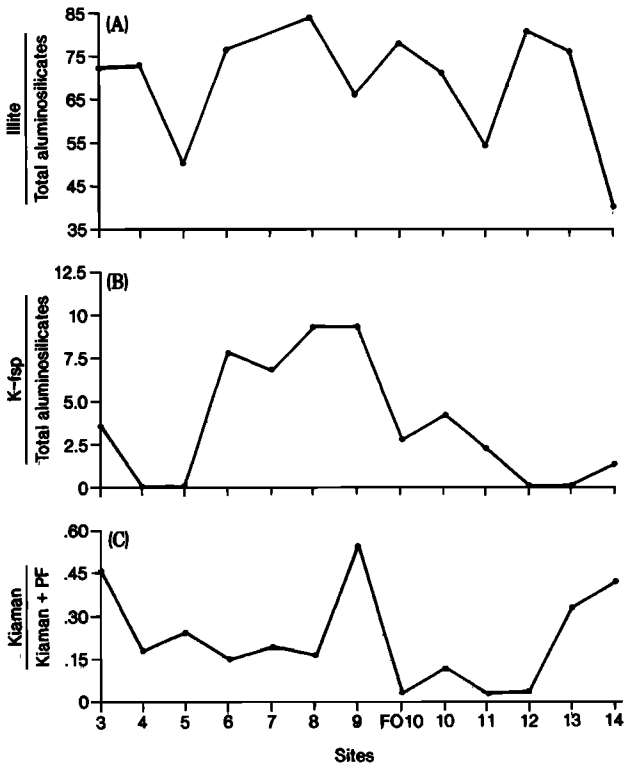


Fig. 10. Comparison of various ratios for remagnetized sites 3-14. (a) Illite/total aluminosilicates. (b) K-feldspar/total aluminosilicates. (c) Kiaman component/total of PF and Kiaman components (i.e., percent "Kiaman magnetite"). The graphs demonstrate the fairly good correlation between Kiaman magnetite and K-feldspar/total aluminosilicates and the poorer correlation between Kiaman magnetite and Illite/total aluminosilicates.

T_{ub} (Figure 6). Since magnetite is the magnetic mineral found at all sites across the transect, it is probably the carrier of the resolvable PF and Kiaman magnetic components.

Interestingly, the unusual north, shallow magnetic directions seen in samples from sites 5, 6, 7, 11, and 14, are found in areas with various combinations of magnetic minerals. Sites 11 and 14 contain magnetite as the dominant magnetic mineral, site 6 has magnetite plus pyrrhotite, and sites 5 and 7 contain magnetite and hematite. It seems likely, therefore, that the north, shallow component is also carried by magnetite.

Concentration and grain size. In order to estimate the relative concentration and grain size distribution of magnetite across the transect, various intensity measurements were made. These include measurement of NRM, IRM, ARM, and X. For IRM intensity, the value at 0.3 T (the saturation value for magnetite) was used for comparison with the other parameters. In a general way, most magnetic intensity averages increase from low values in the western part of the transect to moderate and high values in the central part of the transect and then to low values again in the east (Figure 3).

The sizes of the Kiaman components across the transect correlate fairly well with the magnetite concentration as estimated using X (Figure 6). Both parameters have the highest values in the central part of the transect and lower values to the west and east. This demonstrates that more intensely remagnetized areas generally have greater concentrations of magnetite and vice versa. Therefore, the Kiaman remagnetization in the northern Appalachian Basin, like that in the southern Appalachian Basin [McCabe et al., 1989], may be related to the synthesis of new magnetite.

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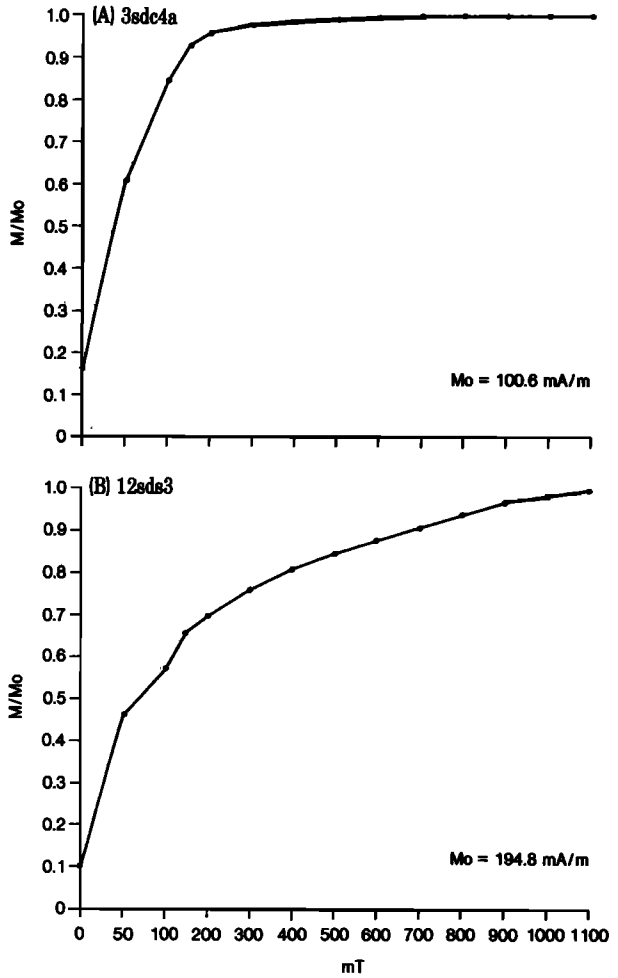


Fig. 11. IRM acquisition graphs. (a) Typical curve from sites which have magnetite as the dominant magnetic mineral (i.e., sites 1, 2, 3, 4, 8, 9, FO10, 10, 11, 13, and 14). Saturation occurs by 0.3 T. (b) Typical curve from sites which have both magnetite and hematite as magnetic carriers (i.e., sites 5, 7, and 12). Saturation does not occur by 1.1 T.

NRM, IRM, and ARM intensities correlate with each other quite well along the transect. X, however, exhibits some discrepancies when compared to the other intensity measurements (Figure 3). For example, sites 3, 9, and 13 have relatively high values for NRM, IRM, and ARM compared with their values for X (Figure 3). A graph of ARM/X (Figure 6) has peaks at sites 3, 9, and 13, which demonstrates these intensity differences. Conversely, site 14 has relatively low values for NRM, IRM, and ARM and a relatively high value for X (Figure 3). These discrepancies among intensity values probably relate to grain size and rock mineralogy. Banerjee et al. [1981] found that the ratio of ARM/X is lowest in coarse-grained magnetite and increases gradually for smaller grain sizes. King et al. [1982] demonstrate that ARM is especially sensitive to fine-grained magnetite, while X is more sensitive to coarse-grained magnetite. These findings are further supported by laboratory results which show that large multidomain grains may have greater X than single-domain and pseudosingle domain magnetite by as much as a factor of 2 [Dunlop, 1986; Heider et al., 1987]. Results from this transect, described above, show relatively high ratios of ARM/X for sites 3, 9, and 13. This probably reflects comparatively large populations of fine-grained magnetite at these sites. This fine-grained magnetite is most likely responsible for the very well-resolved Kiaman directions at sites 3, 9, and 13 (Figure 5). In general, the size of the Kiaman components correlate well with the ARM/X ratios at the sites sampled (Figure 6). Assuming that a higher ARM/X ratio indicates a greater percentage of small grains, this seems to provide evidence that, in general, the size of the Kiaman component at a site is related to the concentration of small grains at the site. This observation, combined with the correlation between magnetite concentration and Kiaman components described above, demonstrates that authigenic formation of new, small magnetite grains may be responsible for the Kiaman remagnetization in the northern Appalachian Basin. These findings support those reported by Jackson [1990], who suggests that although coarse-grained magnetite is common in these rocks, most of the remanence is carried by fine-grained magnetite.

In contrast to sites 3, 9, and 13, site 14 has high values for X and a low ARM/X ratio (Figure 6). Site 14, therefore, may have a relatively large population of multidomain magnetite grains. This might explain the poorly resolved Kiaman components at this site. Another possible explanation for the high X at site 14 is the great concentration of clay at this site. Bulk susceptibility can be dominated by paramagnetic clay minerals when the concentration of ferromagnetic minerals is low [Lowrie and Heller, 1982].

Relevant to the above discussion is the fact that sites 5, 11, and 14, all of which have well-resolved PF directions but poorly resolved (or unresolved) Kiaman directions, all have very low values for ARM/X (Figure 6). It seems likely that these sites have large populations of multidomain grains. In fact, the size of the PF components correlate fairly well with X across the transect (Figure 12). These results suggest that the average size of the PF component at a site is related to the concentration of large magnetite grains.

Unblocking temperature. In general, the sizes of the Kiaman components correlate with the maximum T_{ub} of the Kiaman components across the transect. That is, in those sites with larger Kiaman components, the Kiaman components unblock at higher average temperatures and vice versa. In addition, higher T_{ub} correlate with larger ARM/X ratios and lower T_{ub} correlate with smaller ARM/X ratios (Figure 6). These results indicate that the sizes and T_{ub} of the Kiaman components are associated with average grain size of magnetite.

Magnetic extracts. The magnetic extracts from the sites sampled exhibit some variability. Extracts from all of the sites contain irregularly shaped individual particles and/or aggregates of irregular particles. In addition, some sites contain euhedral magnetite grains

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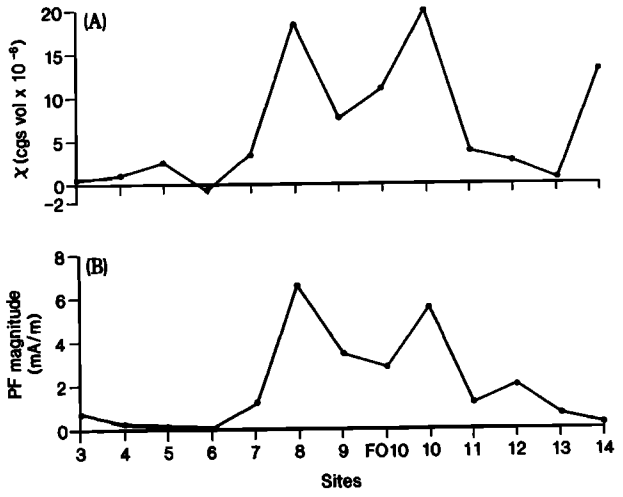


Fig. 12. Comparison of average values for (a) X and (b) average sizes of PF components in remagnetized sites 3-14, showing the correlation between these parameters.

and/or magnetic spheres (Table 2 and Figure 8). The irregular grains are by far the most abundant magnetic particles at all of the sites studied. Euhedral particles and spheres, when present, make up a very small percentage of the grains at any particular site. It must be pointed out that analyses of magnetic extracts cannot be considered quantitative because of the alterations which might occur during the extraction procedure. For example, the rock crushing prior to extraction might damage some of the magnetic particles. However, extensive SEM observations of the extracts from our transect did not reveal broken pieces of magnetite spheres or broken euhedral grains; this leads to the conclusion that these types of grains are indeed relatively uncommon in the rocks collected for this study.

The composition of the grains in the magnetic extracts is somewhat variable. EDAX analyses demonstrate that about two-thirds of the irregular particles contain Fe as the only cation while the other one-third contain Fe plus some combination of Cr, Ni, Ti, Mn, and Zn and sometimes ions such as Ca, Al, Si, and Mg. The presence of the Ca, Al, Si, and Mg group of ions may be due to clays trapped in ferric oxide coatings on some magnetic particles [Suttill et al., 1982]. Some of the extracted grains contain no Fe. Their shape and composition correspond with that of spinel. The irregular particles measured range in size from 6-120 microns (Table 2).

The euhedral particles tested with the EDAX contain Fe as the only cation. In this respect they differ from euhedral grains from unremagnetized site 2, some of which contain Ti and Cr ions in addition to Fe. Euhedral grains measured range in size from 10 to 37.5 microns (Table 2).

The magnetic spheres also demonstrate some variability. Although about 70% of the spheres observed contain Fe as the only cation, the remaining 30% contain Fe plus one or more of the following ions: Cr, Mn, Ti, Zn, Al, Si, Ca. A few spheres appear to be pyrite or spinel. The magnetic spheres measured range in size from less than 1 to 46 microns (Table 2). Many researchers believe that the magnetic spheres form diagenetically or authigenically and suggest that these particles are responsible for the Kiaman remagnetization (reviewed by McCabe and Elmore [1989]). SEM observations of the Onondaga Limestone from New York demonstrate that at least some magnetic spheres are derived by alteration and replacement of pyrite framboids [Suk et al., 1990a].

Regardless of how the magnetic spheres form, recent studies indicate that the spheres visible with the SEM are not likely to carry the major part of the Kiaman remagnetization. As a result of detailed rock magnetic studies on three remagnetized carbonate units, including the Onondaga Limestone of New York, Jackson [1990] concluded that although coarse-grained magnetite is common in these rocks, the remanence is principally carried by fine-grained (single-domain) magnetite. Observations made in this study also indicate that the carriers of the Kiaman magnetization are probably small (single-domain?) grains. Evidence for this, discussed previously, is the fact that sites with very strong Kiaman directions have high ARM/X ratios, indicating an abundance of single-domain magnetite grains. Conversely, sites with very poor Kiaman components have low values for ARM/X, indicating a preponderance of large magnetite grains. Sites along the transect with more intermediate values for ARM/X may have a greater mixture of small and large grains.

The studies of Suk et al. [1990a,b] are relevant to the discussion of grain size. In studies of magnetite from Tennessee and from the Onondaga Limestone of New York, Suk et al. found that both non-spherical magnetite grains and magnetite spheres are polycrystalline, possibly containing single-domain or pseudosingle domain subunits. They believe that their results may explain the rock magnetic properties observed by Jackson [1990], which demonstrate that the Kiaman remagnetization is apparently carried by single-domain magnetite. However, even if the magnetite spheres are composed of small subunits, they are apparently not present in sufficient quantities to carry the Kiaman remagnetization in New York. Observation of the magnetic extracts from the present study reveals that magnetite spheres are apparently present in very low concentrations. In fact, the extracts from some sites with well-resolved Kiaman directions appear to contain few or no spheres (e.g., sites 3, 9, and 10). On the other hand, the extract from site 5, which has extremely poorly resolved Kiaman directions, contains a relatively large number of magnetite spheres.

Some of the numerous irregular particles visible with the SEM, if they are composed of single-domain and pseudosingle domain subunits, may be the carriers of the Kiaman magnetic components. Another possibility is that very small grains, incapable of resolution with the SEM, are responsible for the Kiaman remagnetization. The appearance of such fine-grained carriers can only be speculated upon at the present time. Results of rock magnetic analyses, however, led Jackson [1990] to suggest that the single-domain fraction consists of equant grains which may have a diagenetic origin.

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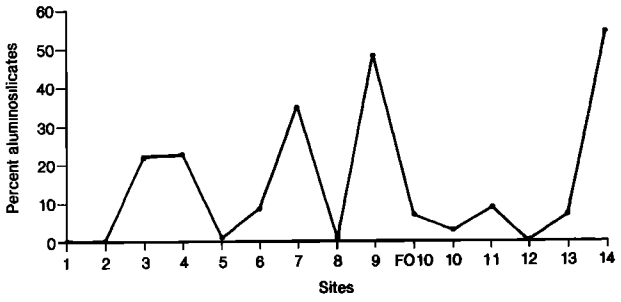


Fig. 13. Graph showing percent total aluminosilicates across the transect demonstrating the variable values at different sites.

An interesting observation concerning the magnetic extracts is that no euhedral particles were observed from site FO10 eastward (Table 2). This area extends from Chittinango to South Bethlehem, New York (Figure 1). If euhedral grains were present in these areas, they may have been subjected to diagenetic changes over time which degraded them.

X Ray diffraction. X ray powder diffraction analyses demonstrate that the percent of total aluminosilicates across the transect is quite irregular (Figure 13). Some sites, such as 1, 2, 5, 8, 10, and 12, have extremely low percentages of these minerals, while other sites, such as 3, 4, 7, and 9, have relatively high concentrations. Site 14, being a clay shale, has a very large proportion of aluminosilicates.

Of greater interest than percent total aluminosilicates is the relative proportion of illite and K-feldspar along the transect. As mentioned previously, there has been speculation concerning a possible relationship between formation of authigenic or diagenetic minerals such as magnetite, K-feldspar and illite in remagnetized carbonates. Some sites along the transect have significant concentrations of illite. This is not surprising, since illite is a widespread clay mineral, being especially abundant in the soils and sediments of eastern North America [Kennett, 1982]. Although illite concentrations are generally significantly higher than K-feldspar concentrations, there is a fairly good correlation between percent K-feldspar and percent illite across the transect (Figure 14). In fact, percent K-feldspar and percent illite both exhibit quite a good correlation with the total concentration of "other" terrigenous aluminosilicates across the study area, supporting a possible detrital origin for most of these minerals.

If there is a connection between synthesis of K-feldspar and/or illite and formation of authigenic magnetite, then these minerals formed at the time of remagnetization, during the Kiaman interval. In fact, there is evidence for late Paleozoic authigenesis of K-feldspar and for K-bentonite illitization in parts of the Appalachian Basin [Elliott and Aronson, 1987; Hearn et al., 1987]. If synthesis of K-feldspar and/or illite are related to magnetite formation in the Ontario-New York study area, the percent K-feldspar and/or percent illite should correlate with the "Kiaman magnetite". To test this, the size of the Kiaman component was compared with the percent K-feldspar and percent illite across the transect (Figure 14); also, the ratio of the Kiaman component/sum of all the magnetic components (which presumably reflects the percentage of Kiaman magnetite) was compared with the ratios of K-feldspar/total aluminosilicates and illite/total aluminosilicates (Figure 10). Using this approach, the size of the Kiaman component and the percent of K-feldspar and illite exhibit a limited correlation across the study area.

In addition to demonstrating a limited correspondence with the size of the Kiaman component, K-feldspar/total aluminosilicates also shows a limited correlation with Kiaman magnetite; illite/total aluminosilicates, however, does not correlate with Kiaman magnetite (Figure 10). It may be that any variation of illite due to diagenetic effects is lost in the noise of the data, since illite of probable detrital origin dominates the aluminosilicate fraction.

Taken as a whole, the results described above indicate that there may be a connection between acquisition of the Kiaman magnetization and formation of other authigenic minerals, particularly K-feldspar. Although there is an imperfect one to one correlation of K-feldspar with parameters reflecting Kiaman magnetite content, both Kiaman magnetite and K-feldspar concentrations are highest in the central sites and lower to the east and west (Figures 10 and 14). This demonstrates that although much of the K-feldspar in these rocks may be of terrigenous origin, some of it may have formed

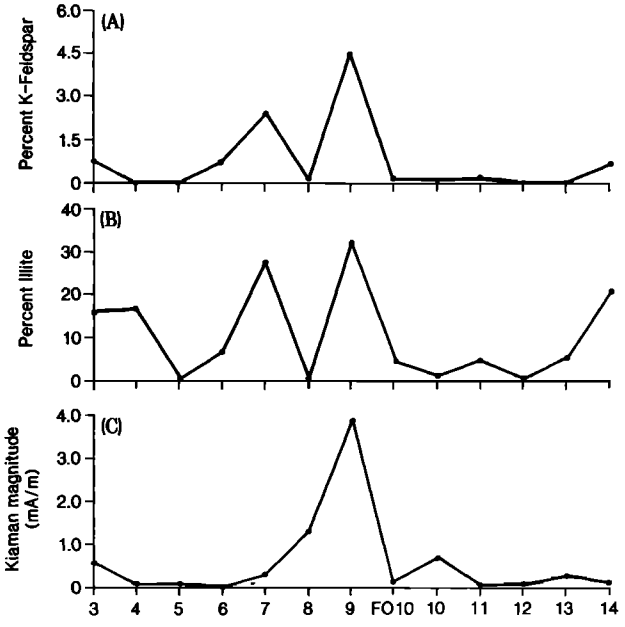


Fig. 14. Comparison of concentrations of aluminosilicates with the sizes of the Kiaman components for remagnetized sites 3-14. (a) Percent K-feldspar. (b) Percent Illite. (c) Average sizes of the Kiaman (south, shallow) components. The graphs demonstrate the strong correlation between the concentrations of illite and K-feldspar and the limited correlation between the sizes of the Kiaman components and the percentages of the aluminosilicates.

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when the rocks were remagnetized during the Kiaman time period.

Discussion and Conclusions

There has been considerable discussion concerning the method by which magnetite-bearing carbonates become remagnetized. The history of this debate and the relevant observations and models were thoroughly reviewed by McCabe and Elmore [1989].

Many of the questions involved in determining how magnetite-bearing carbonates became remagnetized during the Kiaman interval revolve around the size of the magnetic carriers. To briefly summarize this issue, early studies of remagnetized carbonates revealed the presence of multidomain-size magnetite spheres which were interpreted to be of diagenetic or authigenic origin [McCabe et al., 1983; Wisniowiecki et al., 1983]. In addition, it was believed that paleotemperatures in the study areas (e.g., Devonian carbonates of New York) were never sufficiently high to produce thermoviscous remagnetizations in keeping with the predictions of the Pullaiah et al. [1975] blocking temperature-relaxation time relationship for magnetite. McCabe et al. [1983, 1989] therefore suggested that the remagnetization was a fluid-mediated chemical remanent magnetization (CRM). Other studies, however, demonstrated that the Pullaiah et al. [1975] model, formulated for single-domain magnetite, might not be applicable to multidomain magnetite [Kent, 1985; Worm et al., 1988; Enkin and Dunlop, 1988]. This led to the belief that the importance of thermoviscous effects in magnetite-bearing rocks may have been underestimated in low to moderate temperature regimes, such as that in upstate New York.

More recently, Jackson [1990], using detailed rock magnetic studies, found a bimodal distribution of magnetite grain sizes in carbonates (including the Devonian Onondaga Limestone of New York) which carry Kiaman remagnetizations. Jackson suggests that the multidomain particles observed in extracts are responsible for a recent thermoviscous remanent magnetization (TVRM), while previously unreported single-domain magnetites carry a Kiaman CRM.

Results of the present study support Jackson's model. Findings from this study demonstrate that high ARM/X values, which reflect the presence of small magnetite grains [Banerjee et al., 1981; King et al., 1982], correspond with large and well-resolved Kiaman remagnetization components (Figure 6). Likewise, X, which reflects the presence of large magnetite grains [King et al., 1982; Dunlop, 1986; Heider et al., 1987], correlates with the size of PF components across the transect (Figure 12). This suggests that multidomain magnetite carries a PF TVRM and single-domain magnetite carries the Kiaman magnetization, as Jackson proposes.

SEM observations of magnetic extracts from our study area reveal numerous multidomain magnetite grains. These particles have various shapes, sizes, and trace element chemistries (Table 2 and Figure 8). At least some of this magnetite may have formed authigenically during the late Paleozoic remagnetization event. Single-domain magnetite is not resolvable with the observational techniques we used, but its presence in our samples may be inferred from the results of rock magnetic studies discussed previously.

If the Kiaman remagnetizations reside in single-domain magnetite, as suggested by Jackson [1990], the Pullaiah et al. [1975] model should predict the unblocking temperatures with some accuracy. This would support the findings of McCabe et al. [1983], who concluded that the paleotemperatures in the northern Appalachian Basin preclude a thermoviscous remagnetization mechanism.

The presence of north, shallow (Kiaman normal?) directions at sites 5, 6, 7, 11, and 14 of this transect also supports a fluid mediated chemical remagnetization model rather than a thermoviscous one. If the Kiaman Reversed Superchron was interrupted by brief periods of normal polarity, separate fluid pulses during the "reversed" and "normal" periods could have triggered formation of reversed and normal Kiaman components. It is difficult to visualize a thermoviscous regime which could have given rise to this pattern of reversed and normal remagnetizations.

Other data from this study also bolster the argument for a CRM residing in diagenetic or authigenic magnetite. The central part of the transect, which apparently was subjected to the most intensive remagnetization during the Kiaman interval, also has the greatest concentration of magnetite (estimated using X) (Figure 6). These findings demonstrate that more strongly remagnetized areas contain greater amounts of magnetite than less strongly remagnetized areas. In addition, sites 1 and 2, the most weakly magnetic sites of this transect, seem to have escaped late Paleozoic remagnetization. In fact, these sites appear to retain a dual-polarity primary Devonian magnetization (Figure 4). These data coincide with those obtained from the southern Appalachian Basin, where remagnetized carbonates were found to contain more magnetite than unremagnetized carbonates; furthermore, one unremagnetized southern Appalachian site of Mississippian age, with extremely weak magnetism, exhibits dual-polarity Mississippian magnetic directions [McCabe et al., 1989]. These observations lead to the conclusion that remagnetization is associated with the formation of new magnetite in carbonates of both the southern and northern Appalachian Basins. Moreover, although different grain sizes of magnetite may form, it is likely that the Kiaman remagnetization is carried by secondary single-domain magnetite. Carbonates not located along fluid migration pathways would form no secondary magnetite and would remain unremagnetized.

Direct evidence of secondary magnetite in remagnetized carbonates is described by Suk et al. [1990a], who made detailed SEM observations of thin sections of Knox Group carbonates from east Tennessee. They report seeing iron oxide grains in complete contact with secondary dolomite as well as interstitial iron oxide grains and aggregates in association with secondary K-feldspar. Results of the present investigation in the northern Appalachian Basin

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also demonstrate a correlation between K-feldspar and the remagnetization component (Figures 10 and 14). The close association of magnetite with diagenetic minerals such as dolomite and K-feldspar suggests that at least some of the magnetite within remagnetized carbonates is authigenic or diagenetic.

The findings reported here demonstrate a definite association between late Paleozoic remagnetization and formation of secondary magnetite within ancient carbonates of New York. However, a thermal contribution to the remagnetization phenomenon cannot be completely ruled out. As mentioned previously, upstate New York exhibits moderately high thermal maturity, with paleotemperatures ranging from about 100°C near Buffalo to about 200°C near Albany [Friedman and Sanders, 1982; Friedman, 1987; Epstein et al., 1977]. Results of the current study demonstrate that maximum T_{ub} of the Kiaman components generally follow the same trend as the paleotemperatures, showing an overall increase from west to east along the transect. This is especially clear for the T_{ub} of the best resolved Kiaman components, present at sites 3, 9, and 13 (Figures 1 and 6). In addition, the high T_{ub} at these sites correspond with large ARM/X ratios (Figure 6), demonstrating that higher T_{ub} correlate with magnetite of smaller average grain size [Banerjee et al., 1981]. These data lead us to infer that heat is associated with the Kiaman remagnetization; for example, hot fluids may mediate the synthesis of single-domain magnetite particles. McCabe et al. [1989] postulated that remagnetization in the northern Appalachian Basin might have been triggered by migration of heated fluids from the Alleghenian orogenic zone. This model is supported by evidence for migration of high-temperature, high-pressure fluids through Devonian carbonates of the central Appalachians during or after the Alleghenian orogeny [Dorobek, 1989]. The pattern of Kiaman T_{ub} in the northern Appalachian Basin suggests that fluid temperatures were highest near the orogenic zone and decreased toward the Algonquin Arch. This correlates with the paleotemperature trend in the region and is perhap a related phenomenon.

Collectively, the findings presented here support a remagnetization mechanism involving authigenesis of single-domain magnetite. A model which is consistent with the results of this investigation can be envisioned: (1) during the late Paleozoic Alleghenian orogeny, pulses of hot, potassium-rich brines were expelled into the foreland basin; (2) these mediated the synthesis of K-feldspar and authigenic magnetite, including some proportion of single-domain magnetite; and (3) the concentration of authigenic K-feldspar as well as the amounts, sizes, and unblocking temperatures of new magnetite were controlled by a number of parameters. These might include the original mineralogy of the host rock and the compositions, quantities, and temperatures of the migrating fluids.

A better understanding of remagnetization will result from studies concerning the nature of the remagnetizing fluids, the pathways of fluid migration, and the chemical mechanism of magnetite authigenesis.

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References

Bachtadse, V., F. Van der Voo, F.M. Haynes, and S.E. Kesler, Late Paleozoic magnetization of mineralized and unmineralized Ordovician carbonates from east Tennessee: Evidence for a post-ore chemical event, J. Geophys. Res., 92, 14,165-14,176, 1987.

Banerjee, S.K., J. King, and J. Marvin, A rapid method for magnetic granulometry with application to environmental studies, Geophys. Res. Lett., 8, 333-336, 1981.

Channell, J.E.T., R. Freeman, F. Heller, and W. Lowrie, Timing of diagenetic hematite growth in red pelagic limestones from Gubbio (Italy), Earth Planet. Sci. Lett., 58, 189-201, 1982.

Cook, H.E., P.D. Johnson, J.C. Matti, and I. Zemmels, Methods of sample preparation and X-ray diffraction data analysis, X-ray Mineralogy Laboratory, Deep Sea Drilling Project, University of California, Riverside, Initial Rep. Deep Sea Drill. Proj., 28, 83-91, 1975.

Dorobek, S., Migration of orogenic fluids through the Siluro-Devonian Helderberg Group during the late Paleozoic deformation: Constraints on fluid sources and implications for thermal histories of sedimentary basins, Tectonophysics, 159, 24-45, 1989.

Dunlop, D.J., Magnetic mineralogy of unheated and heated red sediments by coercivity spectrum analysis, Geophys. J. R. Astron. Soc., 27, 37-55, 1972.

Dunlop, D.J., Hysteresis properties of magnetite and their dependence on particle size: A test of pseudo-single domain remanence models, J. Geophys. Res., 91, 9569-9584, 1986.

Elliott, W.C., and J.L. Aronson, Alleghenian episode of K-bentonite illitization in the southern Appalachian Basin, Geology, 15, 735-739, 1987.

Enkin, R.J., and D.J. Dunlop, The demagnetization temperature necessary to remove viscous remanent magnetization, Geophys. Res. Lett., 15, 514-517, 1988.

Epstein, A.G., J.B. Epstein, and L.D. Harris, Conodont color alteration-An index to organic metamorphism, U.S. Geol. Surv. Prof. Pap., 995, 27 pp., 1977.

Freeman, R., Magnetic mineralogy of pelagic limestones, Geophys. J. R. Astron. Soc., 85, 433-452, 1986.

Friedman, G.M., Vertical movements of the crust: Case histories from the northern Appalachian Basin, Geology, 15, 1130-1133, 1987.

Friedman, G.M., and J.E. Sanders, Time-temperature-burial significance of Devonian anthracite implies former great (@ 6.5 km) depth of burial of Catskill Mountains, New York, Geology, 10, 93-96, 1982.

21562202b, 1992, B4, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/91JB02746 by Curtin University Library, Wiley Online Library on [05/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Hay, R.L., M. Lee, D.R. Kolata, J.C. Matthews, and J.P. Morton, Episodic potassic diagenesis of Ordovician tuffs in the Mississippi Valley area, Geology, 16, 743-747, 1988.

Hearn, P.P., Jr., J.F. Sutter, and H.E. Belkin, Evidence for Late-Paleozoic brine migration in Cambrian carbonate rocks of the central and southern Appalachians: Implications for Mississippi Valley-type sulfide mineralization, Geochim. Cosmochim. Acta, 51, 1323-1334, 1987.

Heider, F., D.J. Dunlop, and N. Sugiura, Magnetic properties of hydrothermally recrystallized magnetite crystals, Science, 236, 1287-1290, 1987.

Jackson, M., Diagenetic sources of stable remanence in remagnetized Paleozoic cratonic carbonates: A rock magnetic study, J. Geophys. Res., 95, 2753-2761, 1990.

Jackson, M., C. McCabe, M.M. Ballard, and R. Van der Voo, Magnetite authigenesis and diagenetic paleotemperature across the northern Appalachian Basin, Geology, 16, 592-595, 1988.

Johnsson, M.J., Distribution of maximum burial temperatures across the northern Appalachian Basin and implications for Carboniferous sedimentation patterns, Geology, 14, 384-387, 1986.

Kennett, J.P., Marine Geology, 813 pp., Prentice-Hall, Englewood Cliffs, N.J., 1982.

Kent, D.V., Paleomagnetism of the Devonian Onondaga revisited, J. Geophys. Res., 84, 3576-3588, 1979.

Kent, D.V., Thermoviscous remagnetization in some Appalachian Limestones, Geophys. Res. Lett., 12, 805-808, 1985.

Kent, D.V., and J.D. Miller, Redbeds and thermoviscous magnetization theory, Geophys. Res. Lett., 14, 327-330, 1987.

Kent, D.V., and N.D. Opdyke, Multicomponent magnetizations from the Mississippian Mauch Chunk Formation of the Central Appalachians and their tectonic implications, J. Geophys. Res., 90, 5371-5383, 1985.

King, J.W., S.K. Banerjee, J. Marvin, and O. Ozdemir, A comparison of different magnetic methods for determining the relative grain size of magnetite in natural materials: Some results from lake sediments, Earth Planet. Sci. Lett., 59, 404-419, 1982.

Kirschvink, J.L., The least-squares line and plane and the analysis of palaeomagnetic data, Geophys. J. R. Astron. Soc., 62, 699-718, 1980.

Lowrie, W., Identification of ferromagnetic minerals in a rock by coercivity and unblocking temperature properties, Geophys. Res. Lett., 17, 159-162, 1990.

Lowrie, W., and F. Heller, Magnetic properties of marine limestones, Rev. Geophys., 20, 171-192, 1982.

Lu, G., S. Marshak, and D.V. Kent, Characteristics of magnetic carriers responsible for late Paleozoic remagnetization in carbonate strata of the mid-continent, U.S.A., Earth Planet. Sci. Lett., 99, 351-361, 1990.

McCabe, C., and R. D. Elmore, The occurrence and origin of late Paleozoic remagnetization in the sedimentary rocks of North America, Rev. Geophys., 27, 471-494, 1989.

McCabe, C., R. Van der Voo, D.R. Peacor, C.R. Scotese, and R. Freeman, Diagenetic magnetite carries ancient yet secondary remanence in some Paleozoic sedimentary carbonates, Geology, 11, 221-223, 1983.

McCabe, C., R. Van der Voo, and M.M. Ballard, Late Paleozoic remagnetization of the Trenton Limestone, Geophys. Res. Lett., 11, 979-982, 1984.

McCabe, C., M. Jackson, and B. Saffer, Regional patterns of magnetite authigenesis in the Appalachian Basin: Implications for the mechanism of late Paleozoic remagnetization, J. Geophys. Res., 94, 10,429-10,443, 1989.

Miller, J.D., and D.V. Kent, Paleomagnetism of the Upper Devonian Catskill Formation from the southern limb of the Pennsylvania salient: Possible evidence of oroclinal bending, Geophys. Res. Lett., 13, 1173-1176, 1986.

Pullaiah, G., E. Irving, K.L. Buchan, and D.J. Dunlop, Magnetization changes caused by burial and uplift, Earth Planet. Sci. Lett., 28, 133-143, 1975.

Suk, D., D.R. Peacor, and R. Van der Voo, Replacement of pyrite framboids by magnetite in limestone and implications for palaeomagnetism, Nature, 345, 611-613, 1990a.

Suk, D., R. Van der Voo, and D.R. Peacor, Scanning and transmission electron microscope observations of magnetite and other iron phases in Ordovician carbonates from east Tennessee, J. Geophys. Res., 95, 12,327-12,336, 1990b.

Suttill, R.J., P. Turner, and D.J. Vaughan, The geochemistry of iron in recent tidal-flat sediments of the Wash area, England: A mineralogical, Mossbauer, and magnetic study, Geochim. Cosmochim. Acta, 46, 205-217, 1982.

Van der Voo, R., Paleomagnetism of continental North America: The craton, its margins and the Appalachian belt, in Geophysical Framework of the Continental United States, Mem. Geol. Soc. Am., 172, 447-470, 1989.

Wisniowiecki, M.J., R. Van der Voo, C. McCabe, and W.C. Kelly, A Pennsylvanian paleomagnetic pole from the mineralized late Cambrian Bonneterre Formation, southeast Missouri, J. Geophys. Res., 88, 6540-6548, 1983.

Worm, H.-U., M. Jackson, P. Kelso, and S.K. Banerjee, Thermal demagnetization of partial thermoremanent magnetization, J. Geophys. Res., 93, 12,196-12,204, 1988.

Zijderveld, J.D.A., A.C. demagnetization of rocks: Analysis of results, in Methods in Paleomagnetism, pp.245-286, edited by D.W. Collinson and S.K. Runcorn, Elsevier, New York, 1967.

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